

Homework — 2.2.1 Programming Techniques

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 — set after the 2.2.1 lesson. Show traces/working.

Part A — This week's topic (~14 marks)

1. State the difference between **count-controlled** and **condition-controlled** iteration, and name **one** OCR construct used for each. [2] (AO1)

2. A program validates input by re-asking for a value until it is greater than 0. Explain why a **do...until** loop is more suitable here than a **while** loop. [2] (AO1)

3. Trace the call **power(2, 4)** for the recursive function below. Show the value returned at **each** level of the recursion and the final result. [3] (AO2)

```
function power(base, exp)
  if exp = 0 then
    return 1
  else
    return base * power(base, exp - 1)
  endif
endfunction
```

4. Study the program below. Note that in this language an assignment to a same-named variable inside a procedure creates a **new local** variable; it does not change the global.

```
count = 8

procedure reset()
  count = 0
  print(count)
endprocedure

reset()
print(count)
```

(a) State the two values printed, in order. [2] (AO2)

Output line 1: Output line 2:

(b) Explain your answer to part (a), referring to **scope**. [2] (AO2)

5. A subroutine is called with `total = 10`.

```
procedure scaleByVal(byVal n)
  n = n * 2
endprocedure

procedure scaleByRef(byRef n)
  n = n * 2
endprocedure
```

State the value of `total` after each call below and justify each answer. [2] (AO2)

After `scaleByVal(total)` : total = because

After `scaleByRef(total) : total = __` **because** ____

6. Define the term **encapsulation** and state **one** benefit it provides in object-oriented programming. [1]
(AO1)

Part B — Retrieval: keep it fresh (~6 marks)

7. (2.1) State what is meant by **abstraction** and give **one** reason it is useful when solving a problem. [2]

8. (1.4.2) A **stack** and a **queue** are both abstract data structures. State the order in which items are removed from each (use the correct three-letter acronym). [2]

- Stack: ____
- Queue: ____

9. (2.1) Explain what is meant by **decomposition**. [2]